

SURVIVAL ANALYSIS PORTFOLIO

Survival Analysis of Breast Cancer Patients Using Kaplan–Meier and Cox Proportional Hazards Models

Evidence from the METABRIC Dataset

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Analysis conducted in R • survival, survminer, gtsummary, broom, ggplot2

Executive Summary

Breast cancer remains one of the leading causes of cancer-related morbidity and mortality among women worldwide. Accurate estimation of patient survival and identification of prognostic factors are essential for guiding clinical decision-making, risk stratification, and treatment planning. This portfolio presents a complete survival analysis of breast cancer patients using the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) dataset, conducted in R using the Kaplan–Meier estimator, log-rank tests, and Cox proportional hazards regression.

After data preparation and complete-case analysis, 1,406 of the original 2,509 patients were retained for the final analytical dataset. Median overall survival was estimated at 163.2 months (95% CI: 150.6–176.4), with survival probability declining to 60.0% at 120 months and 46.5% at 180 months.

Log-rank tests showed significant survival differences by estrogen receptor (ER) status, HER2 status, and tumor stage. Multivariable Cox regression identified older age, larger tumor size, HER2 positivity, and higher histologic grade as independent predictors of increased mortality, while ER-positive tumors were associated with improved survival after adjustment. The final model achieved a concordance index of 0.664. Schoenfeld residual testing indicated violation of the proportional hazards assumption for all covariates, underscoring the need for cautious interpretation of hazard ratios as average effects over follow-up.

This project demonstrates an end-to-end, reproducible survival-analysis workflow — from data cleaning through diagnostic assessment — appropriate for clinical epidemiology and biostatistical research.

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1. Background

Breast cancer is the most frequently diagnosed cancer among women worldwide and a major contributor to cancer-related mortality. Despite advances in screening, diagnosis, and treatment, survival varies considerably because of differences in tumor biology, stage at diagnosis, molecular characteristics, and treatment response.

Time-to-event outcomes require methods that account for censoring, since many patients remain alive or are lost to follow-up before the event of interest occurs. Survival analysis provides this framework: the Kaplan–Meier estimator describes survival probability over time, the log-rank test compares survival between groups, and the Cox proportional hazards model estimates the independent association between multiple predictors and mortality hazard while adjusting for confounding.

Established prognostic markers in breast cancer include patient age, tumor size, tumor stage, histologic grade, and molecular biomarkers such as estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2). Quantifying how these characteristics influence survival supports individualized treatment planning and risk stratification.

2. Objectives

General Objective

To evaluate overall survival and identify important prognostic factors among breast cancer patients using the METABRIC dataset.

Specific Objectives

- To describe the demographic and clinical characteristics of the study population.
- To estimate overall survival using the Kaplan–Meier method.
- To compare survival distributions across clinically relevant subgroups using log-rank tests.
- To identify independent predictors of mortality using Cox proportional hazards regression.
- To evaluate the proportional hazards assumption and overall model performance.

3. Dataset and Variable Description

Data Source and Study Population

The study used the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) dataset [1], a large, well-characterized breast cancer cohort with long-term clinical follow-up and molecular data. The original dataset contained 2,509 patients. After selecting variables required for the planned survival analyses and applying complete-case analysis, the final analytical dataset consisted of 1,406 patients.

Outcome Definition

The primary outcome was overall survival, measured in months from diagnosis to death or end of follow-up. Patients alive at the end of follow-up were treated as right-censored.

Variables Included in the Analysis

Table 1. Description of Variables Included in the Survival Analysis

Variable	Description	Type
overall_survival_months	Time from diagnosis to death or last follow-up (months)	Continuous
event	Outcome indicator (1 = death, 0 = censored)	Binary
age_at_diagnosis	Patient age at diagnosis (years)	Continuous
tumor_size	Maximum diameter of primary tumor (mm)	Continuous
tumor_stage	Clinical stage at diagnosis (I–IV)	Categorical
er_status	Estrogen receptor status (positive/negative)	Binary
her2_status	HER2 receptor status (positive/negative)	Binary
pr_status	Progesterone receptor status (positive/negative)	Binary
neoplasm_histologic_grade	Degree of tumor cell differentiation	Ordinal

Clinical rationale:

- Age and tumor size: larger, later-detected tumors and older age are generally associated with poorer outcomes.
- Tumor stage: Stage I (localized) through Stage IV (distant metastasis) reflects disease extent.
- ER status: ER-positive tumors are hormone-responsive and generally carry a better prognosis than ER-negative tumors.
- HER2 status: HER2-positive tumors historically show more aggressive growth, though targeted therapy has improved outcomes.
- PR status: PR-positive tumors are typically hormone-responsive with a more favorable prognosis.
- Histologic grade: ranges from Grade 1 (well differentiated) to Grade 3 (poorly differentiated, most aggressive).

Data Cleaning and Preparation

Data preparation involved variable selection, assessment of missingness, construction of the event indicator, and creation of the survival object. Missingness was substantial for several candidate variables — three-gene subtype (29.7%), tumor stage (28.7%), laterality (25.5%), cellularity (23.6%), and surgery type (22.1%). Because the objective was to demonstrate a complete, reproducible survival-analysis workflow, complete-case analysis was adopted for the final modeling dataset, retaining 1,406 of 2,509 patients (56%). This approach simplifies estimation and interpretation but may introduce selection bias if excluded patients differ systematically from those retained; this limitation is revisited in Section 8.

4. Statistical Methods

The general analytic approach followed standard survival-analysis methodology [7].

4.1 Descriptive Analysis

Continuous variables were summarized using means, medians, standard deviations, and ranges; categorical variables were summarized using frequencies and percentages. Missingness was assessed prior to modeling, and a complete-case approach was used for the final analytical dataset.

4.2 Kaplan–Meier Survival Estimation

The Kaplan–Meier method [2] was used to estimate the survival function, accounting for right-censoring. Median survival time and 95% confidence intervals were calculated, and survival probabilities were estimated at 12, 36, 60, 120, and 180 months post-diagnosis.

4.3 Log-Rank Test

The log-rank test [3] was used to compare survival distributions between subgroups defined by ER status, HER2 status, and tumor stage. A statistically significant result indicated evidence of differing survival experiences between groups.

4.4 Cox Proportional Hazards Regression

Univariate Cox models [4] were first fitted for each candidate predictor (age at diagnosis, tumor size, tumor stage, ER status, HER2 status, PR status, and histologic grade) to obtain crude hazard ratios (HRs). A multivariable Cox model was then fitted with the same set of predictors simultaneously to estimate adjusted hazard ratios (AHRs), each with 95% confidence intervals and p-values.

4.5 Model Performance

Predictive performance of the final Cox model was assessed using Harrell's concordance index (C-index) [6], which quantifies the model's ability to correctly rank patients by mortality risk (0.5 = chance, 1.0 = perfect discrimination).

4.6 Proportional Hazards Assumption

The proportional hazards assumption was evaluated using Schoenfeld residuals [5], with both variable-specific tests and a global test. A statistically significant result was interpreted as evidence that a covariate's effect changed over time, i.e., violation of the assumption. Scaled Schoenfeld residual plots were examined to visualize time-dependent patterns.

4.7 Significance Level and Software

All tests were two-sided; $p < 0.05$ was considered statistically significant. All analyses were conducted in R [10] using the survival package [8], survminer [9], tidyverse, gtsummary, broom, and ggplot2.

5. Results

5.1 Study Population

Table 2. Selection of the Study Population

Study Population	Number of Patients
Original METABRIC dataset	2,509
Final complete-case analytical dataset	1,406

Interpretation: The original METABRIC dataset contained 2,509 patients. Following variable selection and complete-case analysis, 1,406 patients (approximately 56%) were retained for the final Kaplan–Meier and Cox proportional hazards analyses.

5.2 Missing Data Assessment

Table 3. Variables with the Highest Proportion of Missing Data

Variable	Missing (%)
Three-gene subtype	29.7
Tumor stage	28.7
Laterality	25.5
Cellularity	23.6
Surgery type	22.1

Interpretation: The highest proportions of missing values were observed for three-gene subtype (29.7%), tumor stage (28.7%), laterality (25.5%), cellularity (23.6%), and surgery type (22.1%). Given this substantial missingness among candidate predictors, complete-case analysis was used to construct the final dataset; this may introduce selection bias if excluded patients differ systematically from those retained.

5.3 Distribution of Survival Status

Table 4. Distribution of Survival Status Before and After Complete-Case Analysis

Survival Status	Original Dataset (n = 2,509)	Complete-Case Dataset (n = 1,406)
Alive / Censored	837	621
Dead / Event	1,144	785

Interpretation: Among the 1,406 patients in the complete-case dataset, 785 (55.8%) experienced death and 621 (44.2%) were right-censored. This relatively high event proportion provides adequate statistical information for estimating survival functions and fitting Cox regression models with reasonable precision.

5.4 Follow-Up Characteristics

Table 5. Summary of Follow-up Time (Months)

Statistic	Follow-up Time (Months)
Minimum	0.10
Median	117.67
Mean	127.26
Maximum	351.00

Interpretation: Patients were followed for a median of 117.67 months (range 0.10–351.00). This long observation window supports estimation of both short- and long-term survival outcomes.

5.5 Baseline Characteristics

Table 6. Baseline Characteristics of Breast Cancer Patients (N = 1,406)

Characteristic	Summary
Mean age at diagnosis (years)	60.45
Mean tumor size (mm)	25.78
ER-positive, n (%)	1,077 (76.6)
ER-negative, n (%)	329 (23.4)
HER2-positive, n (%)	175 (12.4)
HER2-negative, n (%)	1,231 (87.6)
PR-positive, n (%)	728 (51.8)
PR-negative, n (%)	678 (48.2)

Interpretation: Mean age at diagnosis was 60.45 years and mean tumor size was 25.78 mm. Most patients had ER-positive tumors (76.6%), while HER2-positive disease was comparatively uncommon (12.4%); PR status was nearly evenly split. This heterogeneous population is well suited for evaluating the prognostic value of established clinicopathological factors.

5.6 Overall Survival

Figure 1. Kaplan–Meier Curve for Overall Survival

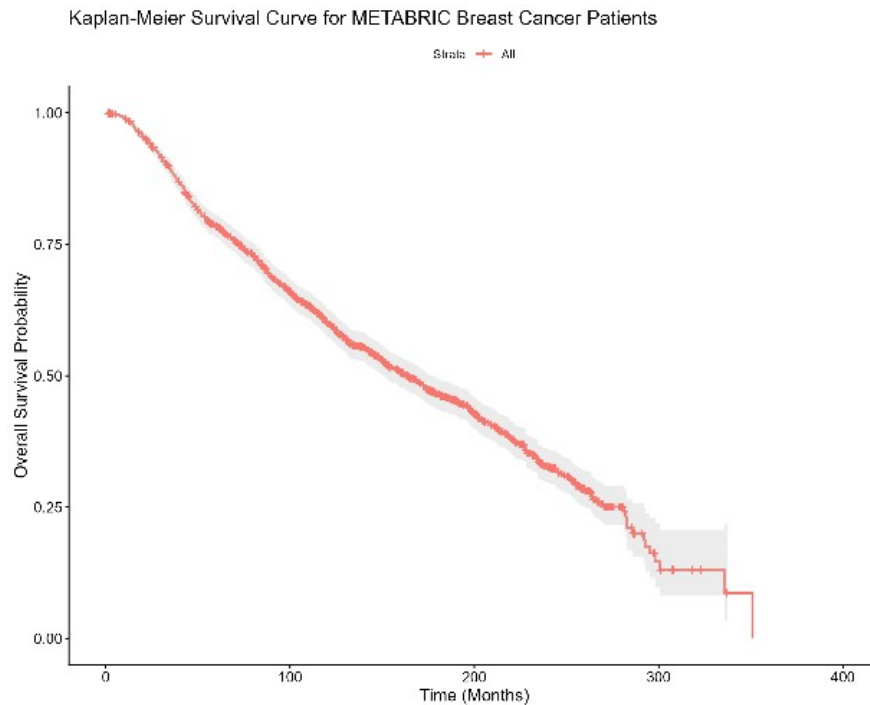


Table 7. Overall Survival Estimate

Survival Measure	Estimate
Median overall survival (months)	163.2
95% Confidence Interval	150.6 – 176.4

Interpretation: Figure 1 presents the Kaplan–Meier estimate of overall survival for the 1,406 breast cancer patients included in the study. The survival curve demonstrates a gradual decline in survival probability throughout the follow-up period, indicating progressive mortality over time.

Survival remained relatively high during the early years following diagnosis but decreased steadily with increasing follow-up. The estimated median overall survival was 163.2 months (95% CI: 150.6–176.4 months), indicating that approximately half of the patients survived for more than 13.6 years after diagnosis.

The shaded region surrounding the survival curve represents the 95% confidence interval of the Kaplan–Meier estimate. Confidence intervals become wider toward the end of follow-up because fewer patients remained under observation, as reflected in the decreasing numbers at risk; this widening reflects increasing statistical uncertainty rather than changes in patient prognosis.

Overall, the Kaplan–Meier curve demonstrates favorable short-term survival but substantial declines in long-term survival among patients with breast cancer, emphasizing the importance of long-term clinical follow-up and prognostic evaluation.

5.7 Survival Probabilities Over Time

Table 8. Kaplan–Meier Estimated Overall Survival Probabilities

Time Since Diagnosis (Months)	Survival Probability (%)
12	98.5
36	88.8

Time Since Diagnosis (Months)	Survival Probability (%)
60	78.3
120	60.0
180	46.5

Interpretation: Survival probability declined progressively over time — from 98.5% at 1 year to 60.0% at 10 years and 46.5% at 15 years — illustrating the sustained long-term mortality risk characteristic of this cohort.

5.8 Survival by Estrogen Receptor Status

Figure 2. Kaplan-Meier Curves by ER Status

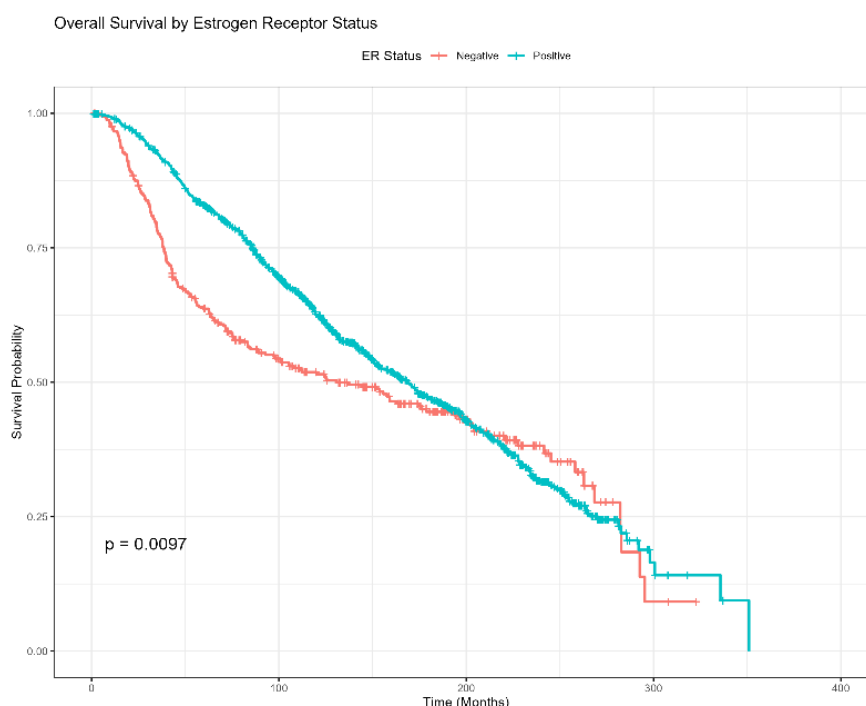


Table 9. Log-Rank Test: Overall Survival by ER Status

Comparison	χ^2 Statistic	p-value
ER-positive vs ER-negative	6.7	0.010

Interpretation: Figure 2 compares overall survival between patients with ER-positive and ER-negative breast tumors. Throughout most of the follow-up period, patients with ER-positive tumors exhibited higher survival probabilities than those with ER-negative tumors, indicating a more favorable prognosis among patients whose tumors expressed estrogen receptors.

The separation between the two survival curves became apparent early during follow-up and remained evident for much of the observation period. The log-rank test demonstrated a statistically significant difference in survival between the two groups ($p = 0.0097$), indicating that the observed survival differences were unlikely to have occurred by chance.

The better survival observed among ER-positive patients is clinically consistent with the established role of estrogen receptor expression as a favorable prognostic marker: ER-positive tumors are generally more

responsive to endocrine therapies, which contribute to improved long-term survival compared with ER-negative disease.

Toward the end of follow-up the curves converged slightly; however, only a small number of patients remained at risk at these later time points, making survival estimates less precise, so greater emphasis should be placed on the earlier and middle portions of follow-up. Overall, the Kaplan–Meier analysis demonstrates that ER-positive breast cancer is associated with significantly better overall survival than ER-negative disease.

5.9 Survival by HER2 Status

Figure 3. Kaplan–Meier Curves by HER2 Status

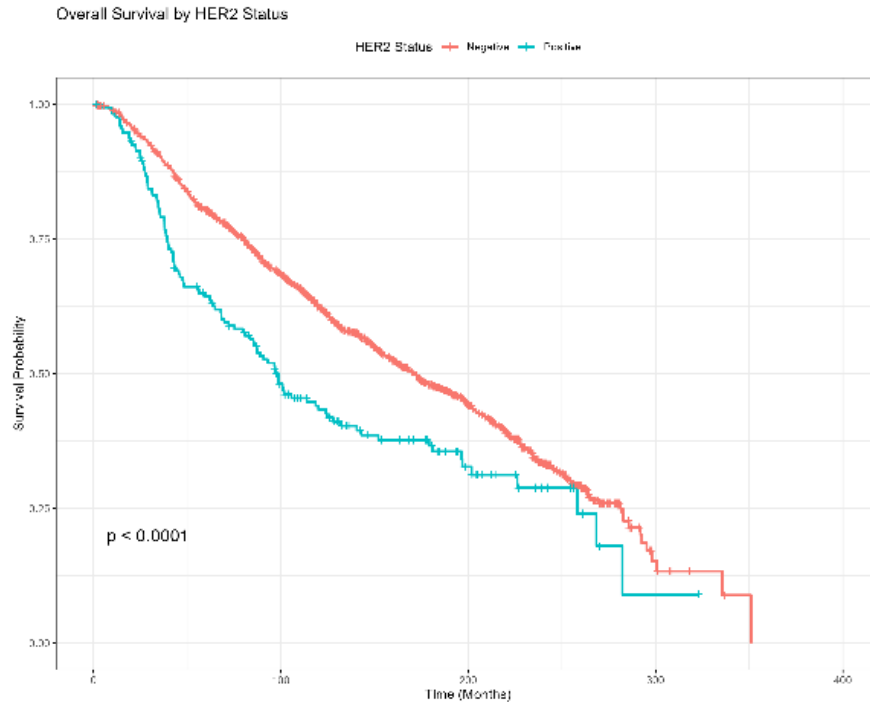


Table 10. Log-Rank Test: Overall Survival by HER2 Status

Comparison	χ^2 Statistic	p-value
HER2-positive vs HER2-negative	18.2	<0.001

Interpretation: Figure 3 illustrates overall survival according to HER2 status. Patients with HER2-positive tumors consistently experienced lower survival probabilities than those with HER2-negative tumors throughout most of the follow-up period.

The separation between the two survival curves occurred relatively early after diagnosis and remained evident across follow-up. The log-rank test confirmed a statistically significant difference in survival between the two groups ($p < 0.0001$), providing strong evidence that survival differed according to HER2 status.

The markedly poorer survival observed among HER2-positive patients is consistent with the recognized aggressive biological behavior of HER2-overexpressing breast cancers. Although targeted anti-HER2 therapies have substantially improved outcomes in modern clinical practice, HER2 positivity remains an important prognostic factor associated with increased mortality risk.

As with the ER survival curves, estimates near the end of follow-up should be interpreted cautiously because relatively few patients remained under observation. Nevertheless, the persistent separation of the curves throughout most of follow-up provides clear visual evidence that HER2 status is strongly associated with overall survival.

5.10 Survival by Tumor Stage

Figure 4. Kaplan–Meier Curves by Tumor Stage

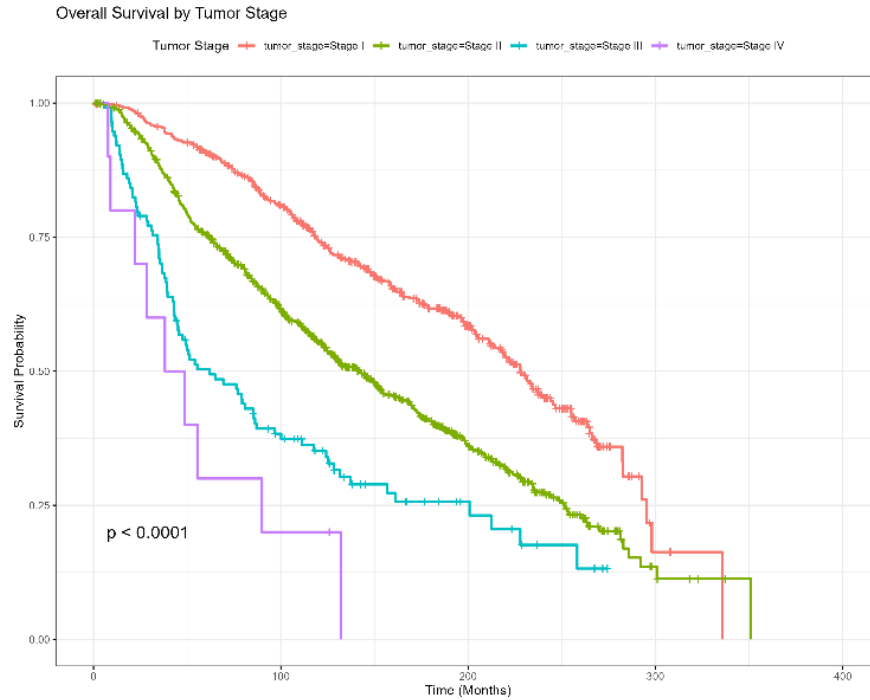


Table 11. Log-Rank Test: Overall Survival by Tumor Stage

Comparison	χ^2 Statistic	p-value
Tumor stage (I–IV)	123.0	<0.001

Interpretation: Figure 4 presents Kaplan–Meier survival curves stratified by tumor stage at diagnosis. Clear and progressive differences in survival were observed across the four disease stages, with survival decreasing as tumor stage increased.

Patients diagnosed with Stage I disease demonstrated the highest survival probabilities throughout follow-up, whereas patients with Stage IV disease experienced the poorest survival. Survival among patients with Stage II disease was intermediate between Stages I and III, while patients with Stage III disease exhibited substantially poorer outcomes than those with earlier-stage disease. This ordered separation of the survival curves closely reflects the expected clinical progression of breast cancer.

The log-rank test demonstrated highly significant differences in survival across tumor stages ($p < 0.0001$), indicating that survival distributions differed substantially among the four groups. Among all subgroup analyses performed in this study, tumor stage exhibited the strongest unadjusted association with overall survival.

The steep decline observed for patients with Stage IV disease highlights the adverse prognosis associated with metastatic breast cancer, whereas the comparatively gradual decline among patients with Stage I disease reflects the substantially better outcomes associated with early diagnosis. Overall, these findings

emphasize tumor stage as one of the most important determinants of survival in unadjusted analyses and reinforce the importance of early detection and timely clinical management of breast cancer.

5.11 Univariate Cox Proportional Hazards Regression

Table 12. Univariate Cox Regression: Factors Associated with Overall Mortality

Predictor	HR	95% CI	p-value
Age (per year)	1.036	1.030–1.040	<0.001
Tumor size (per mm)	1.018	1.015–1.021	<0.001
ER-positive vs ER-negative	0.805	0.683–0.949	0.010
HER2-positive vs HER2-negative	1.546	1.263–1.892	<0.001

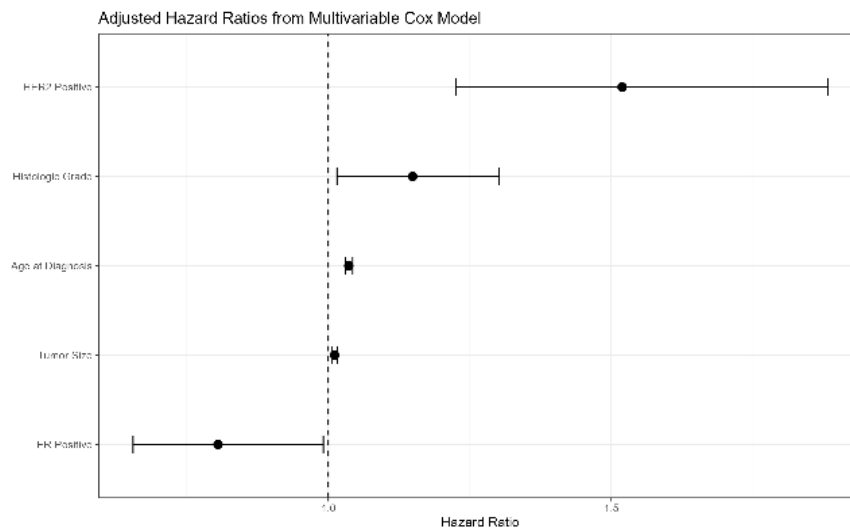
Interpretation: In unadjusted models, each additional year of age increased the mortality hazard by 3.6% (HR = 1.036), and each 1 mm increase in tumor size increased it by 1.8% (HR = 1.018). ER-positive tumors were associated with lower mortality hazard (HR = 0.805), while HER2-positive tumors were associated with substantially higher hazard (HR = 1.546). These crude associations support the clinical relevance of both clinicopathological and molecular predictors prior to adjustment.

5.12 Multivariable Cox Proportional Hazards Regression

Table 13. Multivariable Cox Regression Model for Overall Mortality

Predictor	Adjusted HR	95% CI	p-value
Age (per year)	1.036	1.030–1.043	<0.001
Tumor size (per mm)	1.011	1.007–1.016	<0.001
ER-positive	0.805	0.655–0.991	0.041
HER2-positive	1.519	1.225–1.882	<0.001
Histologic grade	1.149	1.015–1.301	0.028
PR status	Not significant	—	> 0.05
Tumor stage	Not significant	—	> 0.05

Figure 5. Adjusted Hazard Ratios from the Multivariable Cox Model (Forest Plot)



Interpretation: The forest plot summarizes the independent associations between selected clinicopathological factors and overall mortality after adjustment for all other variables in the model. The vertical dashed line at Hazard Ratio (HR) = 1 represents no association with mortality risk; variables with confidence intervals entirely above 1 indicate increased mortality risk, whereas variables with confidence intervals entirely below 1 indicate reduced mortality risk.

The strongest adverse prognostic factor was HER2 positivity (AHR = 1.519, 95% CI: 1.225–1.882), indicating that patients with HER2-positive tumors experienced approximately 52% higher mortality hazard compared with HER2-negative patients after adjustment for other covariates. Histologic grade was also significantly associated with mortality (AHR = 1.149, 95% CI: 1.015–1.301), suggesting that poorly differentiated tumors were associated with worse survival outcomes.

Age at diagnosis remained a strong predictor of mortality (AHR = 1.036 per year, 95% CI: 1.030–1.043), indicating that mortality risk increased progressively with advancing age. Tumor size was independently associated with mortality (AHR = 1.011 per millimeter, 95% CI: 1.007–1.016), demonstrating that larger tumors were associated with poorer survival.

In contrast, ER-positive tumors were associated with improved prognosis (AHR = 0.805, 95% CI: 0.655–0.991), corresponding to approximately 19% lower mortality hazard compared with ER-negative tumors. Overall, the forest plot highlights HER2 status, age, tumor size, histologic grade, and ER status as the most important independent prognostic factors in this cohort, while tumor stage — despite its strong unadjusted association — was attenuated after adjustment, suggesting its prognostic information overlaps with tumor size and histologic grade already in the model.

5.13 Model Performance

Table 14. Predictive Performance of the Final Cox Model

Performance Measure	Value
Concordance Index (C-index)	0.664

Interpretation: The final model achieved a C-index of 0.664, meaning the model correctly ranked relative mortality risk for approximately 66% of randomly selected patient pairs — moderate discrimination, and an acceptable level of performance for a prognostic model built on routinely collected observational data.

5.14 Proportional Hazards Assumption

Table 15. Schoenfeld Residual Test for the Proportional Hazards Assumption

Variable	p-value
Age	<0.001
Tumor stage	<0.001
ER status	<0.001
HER2 status	<0.001
PR status	<0.001
Histologic grade	<0.001
Global test	<0.001

Interpretation: All variable-specific tests and the global test were statistically significant ($p < 0.001$), indicating that one or more covariate effects varied over follow-up time and that the proportional hazards assumption was not fully satisfied. Reported hazard ratios should therefore be interpreted as average effects over the study period rather than strictly time-invariant effects. Future work could explore time-dependent covariates, stratified Cox models, flexible parametric survival models, or accelerated failure time models to better accommodate non-proportional hazards.

6. Discussion

This analysis characterized overall survival and its prognostic determinants among 1,406 breast cancer patients from the METABRIC cohort using complementary survival-analysis methods.

Overall Survival

Median overall survival of 163.2 months, with survival declining to 46.5% by 15 years, reflects the generally favorable but heterogeneous long-term prognosis of breast cancer and underscores the continued mortality risk well beyond the initial years of follow-up.

Age and Tumor Size

Both age (AHR = 1.036 per year) and tumor size (AHR = 1.011 per mm) remained independent predictors after adjustment, consistent with the established clinical understanding that older patients and those with larger, more advanced tumors face greater mortality risk.

Estrogen Receptor (ER) Status

ER-positive tumors were associated with a lower adjusted mortality hazard (AHR = 0.805), consistent with the hormone-responsiveness and generally more favorable biology of ER-positive disease.

HER2 Status

HER2-positive tumors carried a substantially elevated adjusted hazard (AHR = 1.519), affirming HER2 positivity as a marker of more aggressive disease even in a cohort largely predating widespread targeted anti-HER2 therapy.

Histologic Grade

Higher histologic grade was independently associated with increased mortality (AHR = 1.149), consistent with grade reflecting greater cellular abnormality and tumor aggressiveness.

Tumor Stage

Tumor stage showed the strongest unadjusted association with survival (log-rank $\chi^2 = 123.0$) but was not statistically significant after multivariable adjustment. This attenuation likely reflects shared prognostic information with tumor size and histologic grade rather than an absence of clinical relevance, and warrants further investigation of collinearity among staging-related variables.

Model Performance

A concordance index of 0.664 indicates moderate discriminative ability — reasonable for a model built on routinely collected clinical variables, though additional biomarkers or molecular subtyping could likely improve risk stratification.

Proportional Hazards Assumption

Significant Schoenfeld residual tests across all covariates indicate that hazard ratios varied over time. This is a material analytic limitation: the reported hazard ratios should be read as average effects over the observation period, and alternative modeling approaches (time-varying coefficients, stratified Cox models, flexible parametric or accelerated failure time models) merit exploration in future work.

Clinical Implications

Collectively, these findings reinforce the prognostic value of readily available clinicopathological variables — age, tumor size, ER status, HER2 status, and histologic grade — for mortality risk stratification in breast cancer,

while highlighting that tumor stage's prognostic contribution may already be captured by other model variables.

7. Strengths

- Use of a large, well-characterized cohort (METABRIC) with long-term follow-up (up to 351 months).
- A complete, reproducible survival-analysis workflow: data preparation, descriptive analysis, Kaplan-Meier estimation, log-rank testing, univariate and multivariable Cox regression, model performance assessment, and diagnostic testing.
- Formal evaluation of the proportional hazards assumption rather than assuming it holds by default.
- Transparent reporting of missing data and the rationale for the complete-case approach.

8. Limitations

- Complete-case analysis retained only 56% of the original cohort and may introduce selection bias if excluded patients differ systematically from those retained.
- Violation of the proportional hazards assumption across all covariates limits the interpretation of Cox model hazard ratios as constant over time.
- Attenuation of the tumor stage effect after adjustment suggests possible collinearity among staging-related covariates that was not formally assessed (e.g., via variance inflation factors).
- Moderate model discrimination (C-index = 0.664) indicates room for improved risk stratification, potentially through additional molecular or genomic predictors.

9. Conclusions

In this METABRIC-based cohort, age, tumor size, ER status, HER2 status, and histologic grade were independent predictors of overall mortality after adjustment, while tumor stage's strong unadjusted association was attenuated in the multivariable model. The final Cox model demonstrated moderate discriminative performance, and evidence of non-proportional hazards indicates that reported hazard ratios reflect average effects over follow-up rather than constant, time-invariant effects. This project demonstrates a complete, reproducible survival-analysis pipeline suitable for clinical epidemiology and biostatistical research.

10. Recommendations

- Explore alternative modeling strategies — time-dependent covariates, stratified Cox regression, or flexible parametric/accelerated failure time models — to address non-proportional hazards.
- Assess collinearity among tumor stage, tumor size, and histologic grade to clarify tumor stage's independent prognostic contribution.
- Evaluate the impact of complete-case analysis via sensitivity analysis or multiple imputation for variables with substantial missingness.
- Incorporate additional molecular subtyping or genomic variables to improve model discrimination beyond the current C-index of 0.664.

11. References

Vancouver style, numbered in order of appearance.

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12. Appendix

Software and Packages

All analyses were conducted in R [10]. Key packages used: survival [8] (survival object construction, Cox regression), survminer [9] (Kaplan–Meier visualization, log-rank tests), gtsummary and broom (tabulation of model results), tidyverse, janitor, and ggplot2 (data management and visualization).

Appendix A. Main R Code

Note: the code below is a cleaned, de-duplicated version of the analysis script supplied by the author. Exploratory and debugging passes (e.g., repeated re-fits while correcting the tumor stage factor coding) have been removed for clarity and readability; the logic and all variable names are otherwise unchanged from the original script.

Section 1–2. Load Packages and Import Data

```
library(tidyverse); library(janitor); library(survival)
library(survminer); library(ggplot2); library(broom)

metabric <- Breast_Cancer_METABRIC
metabric_raw <- metabric
metabric_clean <- metabric
```

Section 3–4. Data Cleaning and Missing Data Assessment

```
metabric_clean <- metabric_clean %>% clean_names()
metabric_clean <- metabric_clean %>%
  mutate(across(where(is.character), as.factor))

missing_summary <- data.frame(
  Variable = names(metabric_clean),
  Missing = colSums(is.na(metabric_clean)),
  Percent = round(colSums(is.na(metabric_clean)) /
    nrow(metabric_clean) * 100, 2)
)
missing_summary <- missing_summary[order(-missing_summary$Percent), ]
```

Section 5-7. Event Variable and Complete-Case Analytic Dataset

```
metabric_clean <- metabric_clean %>%
  mutate(event = case_when(
    overall_survival_status == "Deceased" ~ 1,
    overall_survival_status == "Living" ~ 0,
    TRUE ~ NA_real_
  ))

surv_data <- metabric_clean %>%
  select(overall_survival_months, event, age_at_diagnosis, tumor_size,
    tumor_stage, er_status, her2_status, pr_status,
    neoplasm_histologic_grade)

surv_data_complete <- surv_data %>% na.omit()
```

Section 8-9. Descriptive Statistics and Survival Object

```
summary(surv_data_complete$overall_survival_months)
table(surv_data_complete$event)
prop.table(table(surv_data_complete$event)) * 100

surv_object <- Surv(
  time = surv_data_complete$overall_survival_months,
  event = surv_data_complete$event
)
```

Section 10-11. Overall Kaplan-Meier Analysis and Figure

```
km_fit <- survfit(surv_object ~ 1, data = surv_data_complete)
summary(km_fit)
surv_median(km_fit)
summary(km_fit, times = c(12, 36, 60, 120, 180))

km_plot <- ggsurvplot(
  km_fit, data = surv_data_complete,
  risk.table = TRUE, conf.int = TRUE,
  xlab = "Time (Months)", ylab = "Overall Survival Probability",
  title = "Kaplan-Meier Survival Curve", palette = "darkblue"
)
ggsave("Figure1_Overall_KM.png", km_plot$plot,
  width = 10, height = 8, dpi = 300)
```

Section 12-13. ER Status — Log-Rank, Cox Model, and Figure

```

km_er <- survfit(surv_object ~ er_status, data = surv_data_complete)
er_logrank <- survdiff(surv_object ~ er_status, data = surv_data_complete)
cox_er <- coxph(surv_object ~ er_status, data = surv_data_complete)
summary(cox_er)

er_plot <- ggsurvplot(km_er, data = surv_data_complete, risk.table = TRUE,
                      pval = TRUE, conf.int = FALSE,
                      legend.title = "ER Status")
ggsave("Figure2_ER_Status.png", er_plot$plot,
        width = 10, height = 8, dpi = 300)

```

Section 14-15. HER2 Status — Log-Rank, Cox Model, and Figure

```

km_her2 <- survfit(surv_object ~ her2_status, data = surv_data_complete)
survdiff(surv_object ~ her2_status, data = surv_data_complete)
cox_her2 <- coxph(surv_object ~ her2_status, data = surv_data_complete)
summary(cox_her2)

her2_plot <- ggsurvplot(km_her2, data = surv_data_complete, risk.table = TRUE,
                       pval = TRUE, conf.int = FALSE,
                       legend.title = "HER2 Status")
ggsave("Figure3_HER2_Status.png", her2_plot$plot,
        width = 10, height = 8, dpi = 300)

```

Section 16-17. Tumor Stage — Log-Rank and Figure

```

surv_data_complete$tumor_stage <- factor(
  surv_data_complete$tumor_stage, levels = c(1, 2, 3, 4),
  labels = c("Stage I", "Stage II", "Stage III", "Stage IV")
)

km_stage <- survfit(surv_object ~ tumor_stage, data = surv_data_complete)
survdiff(surv_object ~ tumor_stage, data = surv_data_complete)

stage_plot <- ggsurvplot(km_stage, data = surv_data_complete, risk.table = TRUE,
                        pval = TRUE, conf.int = FALSE,
                        legend.title = "Tumor Stage")
ggsave("Figure4_Tumor_Stage.png", stage_plot$plot,
        width = 10, height = 8, dpi = 300)

```

Section 18-19. Univariate and Multivariable Cox Regression

```

cox_age <- coxph(surv_object ~ age_at_diagnosis, data = surv_data_complete)
cox_tumor_size <- coxph(surv_object ~ tumor_size, data = surv_data_complete)
cox_stage <- coxph(surv_object ~ tumor_stage, data = surv_data_complete)
summary(cox_age); summary(cox_tumor_size); summary(cox_stage)

cox_multi <- coxph(
  surv_object ~ age_at_diagnosis + tumor_size + tumor_stage +
  er_status + her2_status + pr_status + neoplasm_histologic_grade,
  data = surv_data_complete
)
summary(cox_multi)

```

Section 20–22. PH Diagnostics, Forest Plot, and Model Performance

```

ph_test <- cox.zph(cox_multi)
ph_test
plot(ph_test)

forest_data <- data.frame(
  Variable = c("Age at Diagnosis", "Tumor Size", "ER Positive",
               "HER2 Positive", "Histologic Grade"),
  HR       = c(1.036, 1.011, 0.805, 1.519, 1.149),
  Lower    = c(1.030, 1.007, 0.655, 1.225, 1.015),
  Upper    = c(1.043, 1.016, 0.991, 1.882, 1.301)
)

forest_plot <- ggplot(forest_data, aes(x = HR, y = reorder(Variable, HR))) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = Lower, xmax = Upper), height = 0.2) +
  geom_vline(xintercept = 1, linetype = "dashed") +
  theme_bw()
ggsave("Figure5_Forest_Plot.png", forest_plot,
       width = 10, height = 6, dpi = 300)

summary(cox_multi)$concordance      # model performance (C-index)

```

Reproducibility

Full analysis code: <https://github.com/amdemeskel-biostat/breast-cancer-survival-analysis>